B.5 Continuous Review

L1 Deductibles — Calculus Approach

Eq. Loss $X \longrightarrow f(x) = 0.02x \quad 0 < x < 10.$
Find 10^{th} & 90^{th} percentile for X.

$$P(X \leq x) = \int_0^x 0.02t \, dt$$

$$= 0.01x^2$$

$$P(X \leq s) = 0.1$$
$$0.01s^2 = 0.1$$
$$s = 3.16$$
$$0.01t^2 = 0.9$$
$$t = 9.49$$

Eq. Loss $f(x) = 0.02x \quad 0 < x < 10. \quad d = 4$
$E[\text{Payment}] = ?$

$$\hookrightarrow Y = X - 4 \qquad X > 4$$
$$Y = 0 \qquad X \leq 4$$

$$E[Y] = \int_0^4 0 \cdot f_X(x)\,dx + \int_4^{10} (x-4) \cdot f_X(x)\,dx$$

$$= 0 + \int_4^{10} (x-4)(0.02x)\,dx$$

$$= \int_4^{10} 0.02x^2 - 0.08x \, dx$$

$$= 2.88$$

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L2 Deductibles: Cases Approach

- Let X be a loss. d, then payment Y.

$$Y = \begin{cases} 0 & X \le d \\ X-d & X > d \end{cases}$$

$$E[Y] = E[Y \mid X \le d] \cdot P(X \le d) + E[Y \mid X > d] \cdot P(X > d)$$
$$= 0 + E[X-d \mid X > d] \cdot P(X > d)$$

If $X \sim U(0, a)$

then, $(X-d \mid X > d) \sim U(0, a-d)$
$$E[X-d \mid X > d] = \frac{a-d}{2}$$

If $X \sim \exp(\theta)$
then by memoryless property,

$$E[X-d \mid X > d] = \theta$$

Sometimes, depending on the distribu^n it is easier.

L3 Continuous Review

$$F(x) = P(X \le x)$$
$$F(\infty) = 1 \quad F(-\infty) = 0$$

- If X is continuous, then $F(X)$ is continuous

$$f(x) = \frac{d}{dx} F(x)$$

$$P(a < X \leq b) = \int_a^b f(x)\,dx = F(b) - F(a)$$

- If X is mixed, then $F(x)$ has jumps. Jump size determines probabilities

$$P(X = a) = F(a) - \lim_{x \uparrow a} F(x)$$

- $E[X] = \int x \cdot f(x)\,dx$

  $E[X^2] = \int x^2 \cdot f(x)\,dx$

- $X_1, X_2 \ldots X_n$ are iid

  $\mu = E[X_i]$, $\sigma^2 = Var(X_i)$

  $S = X_1 + X_2 \ldots + X_n$

  CLT,

$$\frac{S - E[S]}{\sqrt{Var(S)}} = \frac{S - n\mu}{\sigma\sqrt{n}} \approx N(0,1)$$

- Uniform(a, b)

$$f(x) = \frac{1}{b-a}, \quad F(x) = \frac{x-a}{b-a}$$

$$E[X] = \frac{a+b}{2}, \quad Var(X) = \frac{(b-a)^2}{12}$$

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- Exponential (θ)

$$f(x) = \frac{1}{\theta} e^{-x/\theta} \quad , \quad F(x) = 1 - e^{-x/\theta}$$

$$E[X] = \theta \quad , \quad Var(X) = \theta^2$$

- Gamma ($\alpha = 2$)

$$f(x) = \frac{x}{\theta^2} e^{-x/\theta} \quad , \quad F(x) = 1 - e^{-x/\theta} - \frac{x}{\theta} e^{-x/\theta}$$

$$E[X] = \alpha\theta \quad , \quad Var(X) = \alpha\theta^2$$